MAHATMA GANDHI UNIVERSITY, KOTTAYAM
MGU-UGP (HONOURS)
THIRD SEMESTER EXAMINATION
(2024 ADMISION ONWARDS)
MG3DSCMAT200-PERSPECTIVES OF MATHEMATICS

Duration: 2 hrs Maximum Marks: 70

Students should attempt atleast one question from each course outcome to enhance their overall outcome attainability.

Part A
Answer any five questions. Each question carries 2 marks

1. Find a polar equation for the circle $x^2 + (y-3)^2 = 9$. [A][1]
2. Find the focus and directrix of the parabola $y^2 = -2x$. [An][2]
3. If α, β, γ are the roots of the equation $27x^3 + 42x^2 - 28x - 8 = 0$, find the values of $\alpha + \beta + \gamma$ and $\alpha\beta\gamma$. [U][3]
4. State Descartes Rule of signs. [A][4]
5. Evaluate $\int_0^1 \int_y^{\sqrt{y}} dxdy$. [A][5]
6. Find the average value of $f(x, y) = sin(x + y)$ over the rectangle $0 \leq x \leq \pi;\ 0 \leq y \leq \pi$. [A][5]
7. Evaluate $\int_0^1 \int_0^1 \int_0^1 (x^2 + y^2 + z^2) dzdydx$. [A][6]
8. Express the spherical coordinate (ρ, ϕ, θ) in terms of rectangular coordinates (x, y, z). [A][7]

Part B
Answer any five questions. Each question carries 6 marks

9. Find the eccentricity of the hyperbola $x^2 - y^2 = 1$. Then find the graph of the hyperbola's foci and directrices. [An][2]
10. Find the cartesian equation for the hyperbola centered at the origin that has a focus at $(3, 0)$ and the line $x = 1$ as the corresponding directrix. [An][2]
11. Solve $2x^3 + x^2 - 7x - 6 = 0$, given that the difference between two of the roots is 3. [A][4]

12. If α, β, γ are the roots of the equation $x^3 + qx + r = 0$, form the equation whose roots are $\beta + \gamma - 2\alpha, \gamma + \alpha - 2\beta, \alpha + \beta - 2\gamma$. [U][3]

13. Sketch the region bounded by the coordinate axes and the line $x + y = 2$. Then express the region's area as double integral and evaluate the integral. [A][5]

14. Sketch the region of integration and write an equivalent double integral of $\int_0^2 \int_{x^2}^{2x} (4x + 3) dy dx$. [A][5]

15. Evaluate the spherical integral $\int_0^{\pi} \int_0^{\pi} \int_0^{2sin\phi} \rho^2 sin\phi \; d\rho d\phi d\theta$. [A][7]

16. Solve the system $u = 3x + 2y, v = x + 4y$ for x & y in terms of u & v. Then find the value of the Jacobian $\frac{\partial(x,y)}{\partial(u,v)}$. [A][7]

Part C
Answer any three questions. Each question carries 10 marks

17. (a) Graph the set points whose polar coordinates satisfy the inequality $0 \leq \theta \leq \frac{\pi}{6}$ and $r \geq 0$. [A][1]

    (b) Show that the equation $x^2 - 4y^2 + 2x + 8y - 7 = 0$ represents a hyperbola. Find its center, asymptotes and foci. [An][2]

18. If $\alpha_1, \alpha_2, ..., \alpha_n$ are the roots of the equation $x^n + p_1 x^{n-1} + p_2 x^{n-2} + ... + p_n = 0$, then the sum of the products of $\alpha_1, \alpha_2, ..., \alpha_n$ taken one, two,.... n at a time, are respectively equal to $-p_1, p_2, -p_3, ..., (-1)^n p_n$. [U][3]

19. (a) Determine the nature of the roots of the equation $x^4 + 3x^2 + 2x - 7 = 0$. [A][4]

    (b) Find the rational roots of the equation $3x^3 - 2x^2 - 6x + 4 = 0$.[U][3]

20. (a) Find the area of the region R bounded by $y = 2x^2$ and $y^2 = 4x$.

    (b) Evaluate $\iint\limits_R e^{x^2+y^2} \, dy dx$, where R is the semicircular region bounded by the $x-$ axis and the curve $y = \sqrt{1 - x^2}$. [A][5]

21. (a) Evaluate $\int_0^1 \int_0^{1-x^2} \int_0^{4-x^2-y} x dz dy dx$. [A][6]

    (b) Evaluate the cylinderical coordinate integral $\int_0^{2\pi} \int_0^1 \int_r^{\sqrt{2-r^2}} dz \; r dr d\theta$. [A][6]

22. (a) Define the Jacobian $\frac{\partial(x,y,z)}{\partial(u,v,w)}$. [A][7]

    (b) Evaluate $\int_0^4 \int_{x=\frac{y}{2}}^{y=\frac{y}{2}+1} \frac{2x-y}{2} \; dx dy$ by applying the transformation $u = \frac{2x-y}{2}$, $v = \frac{y}{2}$ and integrating over an appropriate region in the $uv-$ plane. [A][7]